AI Challenges in Intellectual Property

Part 1

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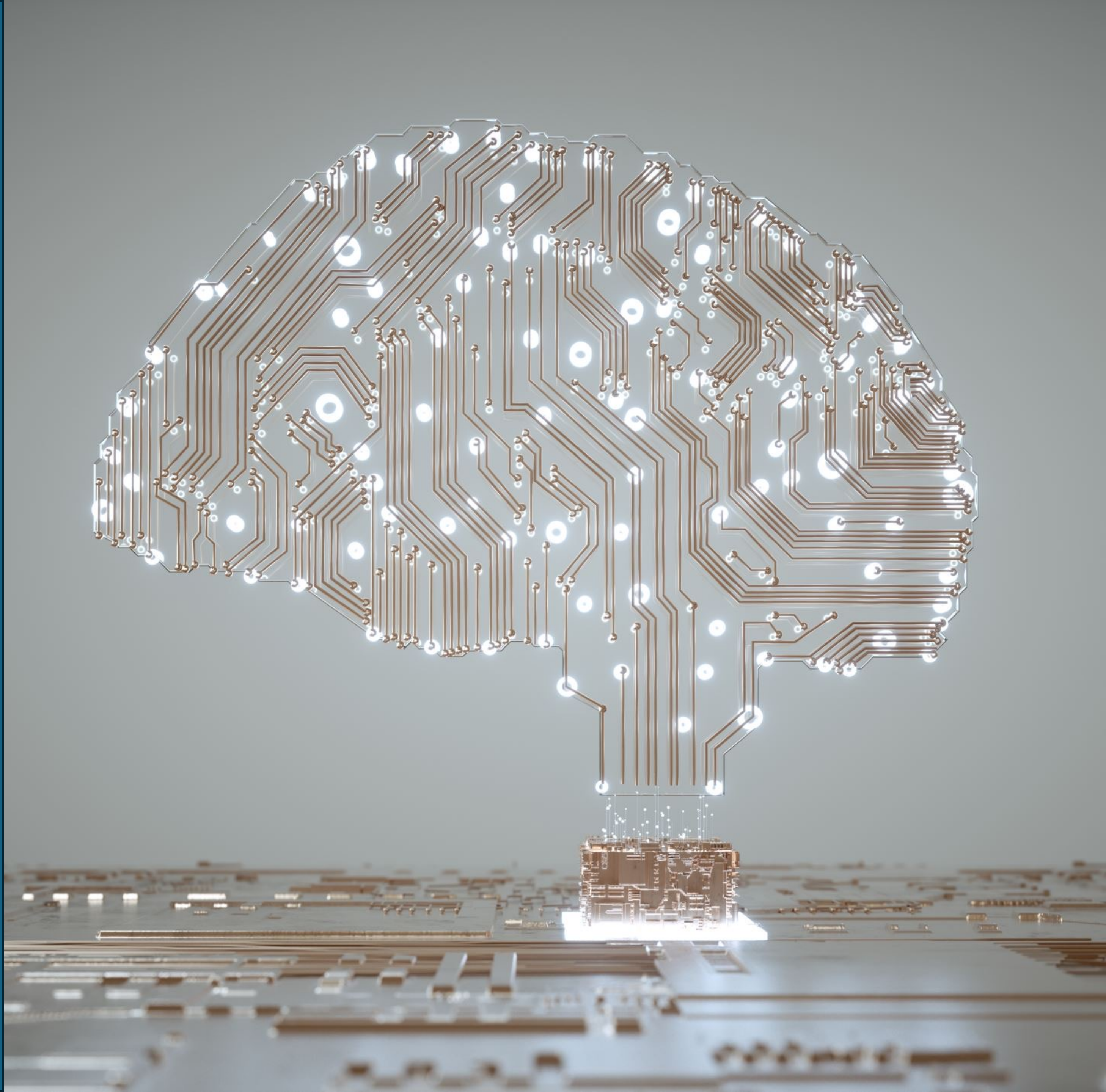


Overview: Intellectual Property (IP)

- What are the primary types of Intellectual property (IP), especially ones relevant to artificial intelligence (AI)?
- Whom do intellectual property (IP) regimes protect? Why are IP laws important?
- Are IP rights transferrable? What is the open-source approach to IP? What is the current status of open-source AI development?

What is Intellectual Property (IP)?

According to WIPO (World Intellectual Property Organization):

- Creations of the “mind” *(Intellectual)*, such as

→ **Inventions** **patent**

Patent: A Primary Type of IP

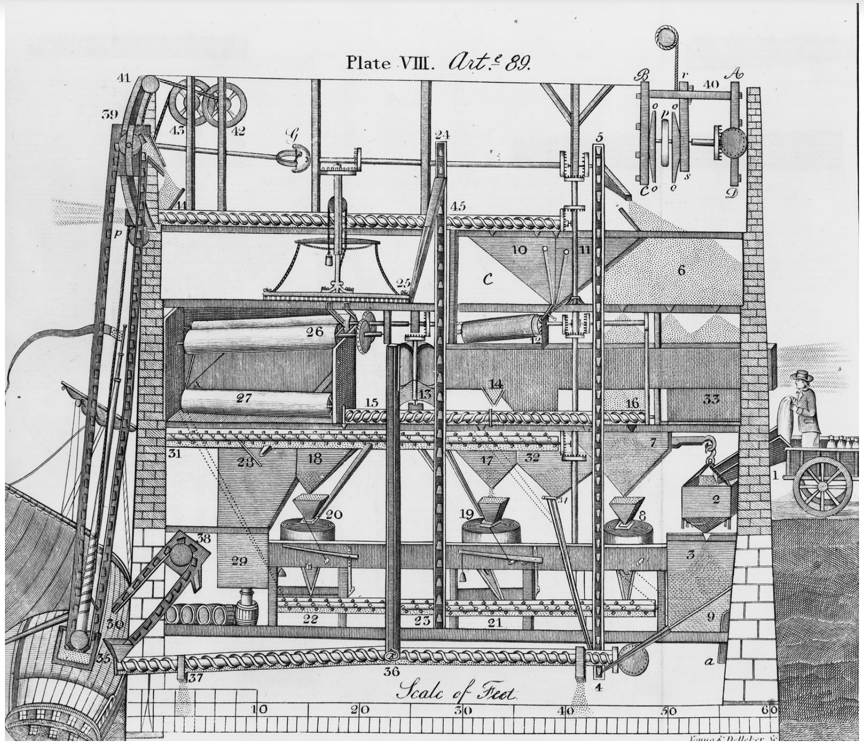
- **Patent**: an exclusive right granted for *an invention*

- To obtain a patent, the invention

→ must be "novel"

→ must involve an inventive step

→ must be industrially applicable, i.e., solves a problem using technology



Automatic gristmill patent, 1790 (granted to Oliver Evans)

What is Intellectual Property (IP)?

According to WIPO (World Intellectual Property Organization):

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→ Inventions

patent

→ **Literary and artistic works, Designs**

copyright

Copyright: A Primary Type of IP

- **Copyright:** *author's* rights to their own created *original works*

- Requirements:

→ must be **original**

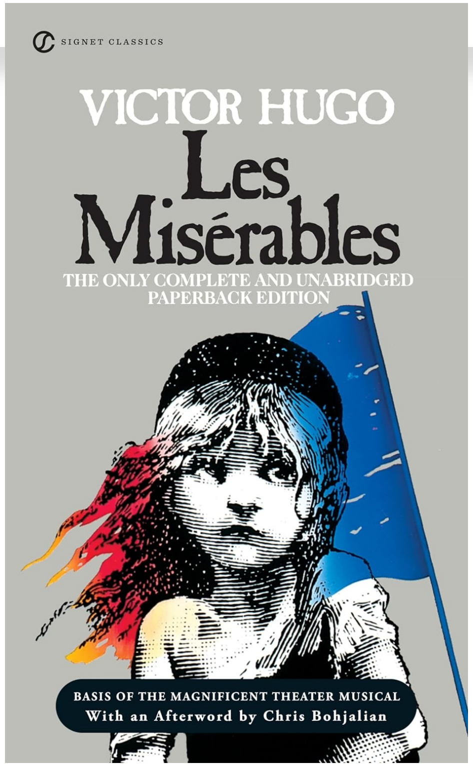
→ must have an **author** (e.g., not accidentally discovered)

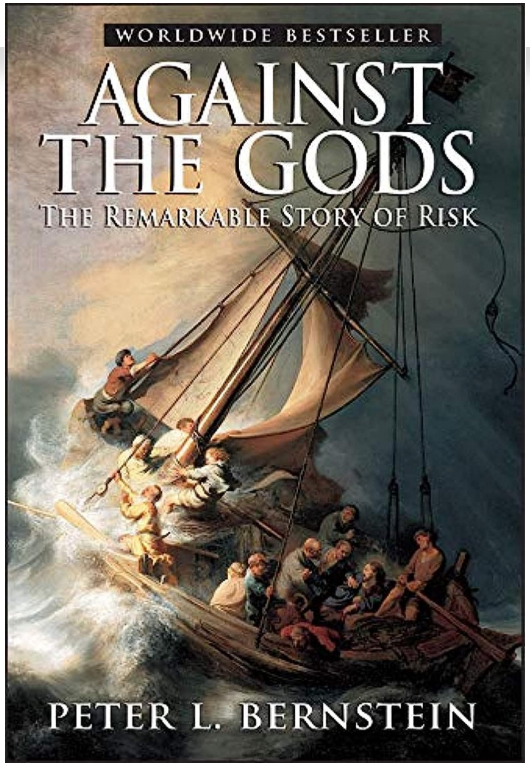
→ minimum level of **creativity**

→ must be fixed in a **tangible medium** (e.g., on a computer screen)

Copyright: A Primary Type of IP

- Covers literary and artistic works, and designs
- Protects “original works of authorship”





Acquiring and Protecting Copyright

How do authors acquire and protect their copyright?

- Copyright is automatically acquired; author does not need to do anything to obtain copyright

- May need to register with Registrar of Copyrights (in the US) to sue others for infringing on your copyright

What is Intellectual Property (IP)?

According to WIPO (World Intellectual Property Organization):

- Creations of the “mind” *(Intellectual)*, such as

→ Inventions patent

→ Literary and artistic works, Designs copyright

→ **Symbols, names and images used in commerce**

trademark

Trademark: A Primary Type of IP

Trademarks:

Distinctive “marks” (e.g., word, phrase, logo) used by producers to distinguish their goods from those produced by other firms

Trademark: A Primary Type of IP

Legal standards for trademarks:

→ must be *distinctive* (i.e., generate a unique association between the mark and the underlying company)

→ must be *in commercial use*

Trademark: A Primary Type of IP

Who owns the trademark?

→ in USA: first one to use it commercially

→ in Europe: needs to be registered and whoever files the registration first

What is Intellectual Property (IP)?

According to WIPO (World Intellectual Property Organization):

- Creations of the "mind" *(Intellectual)*, such as

→ Inventions

→ Literary and artistic works, Designs

→ Symbols, names and images used in commerce

Intangible assets *(property)*, whose owners are granted certain rights and protection similarly to physical properties

Why Protect Intellectual Property (IP)?

"Inventions then cannot in nature be a subject of property. Society may give an exclusive right to the profits arising from them as an encouragement to men to pursue ideas which may produce utility." – Thomas Jefferson to Isaac McPherson (13 August 1813)

"By establishing a marketable right to the use of one's expression, copyright supplies the economic incentive to create and disseminate ideas." – Harper & Row, Publishers, Inc. v. Nation Enterprises, 471 U.S. 539, 558 (1985)

"The copyright is not an inevitable, divine, or natural right that confers on authors the absolute ownership of their creations. It is designed rather to stimulate activity and progress in the arts for the intellectual enrichment of the public." – Pierre N. Leval in "Toward a Fair Use Standard" (1990)

Why Protect Intellectual Property (IP)?

➢ Intellectual Property (IP) is a <u>public good</u>

A public good is…

- ***non-rival:*** one person using it does not prevent another from using it
- ***non-excludable:*** you cannot easily stop anyone from using it easily

Under-provision of a public good is a type of *market failure*

- <u>The private market (creators, firms) do not have the incentive to provide a public good on their own</u>

Why Protect Intellectual Property (IP)?

- To *incentive* something of societal value – e.g., inventions (patents) and creative works of expression (copyright) that would otherwise be ***under-produced in the absence of such laws***

Market failure !

Why Protect Intellectual Property (IP)?

Why don't creators produce inventions or works of art anyway?

- They are costly to produce (e.g., money and time), and require a lot of resources
- It is difficult to prevent others from copying or using them (without compensating the original creator) once they exist
- There is no economic returns to the creators themselves

Why Are IP Laws Important?

- For creators
- For private businesses
- For society at large

IP Laws Protect Creators

- IP laws provide **exclusive rights to the creator(s)** of an intellectual property (IP)

Creators: Inventors, & Authors of artistic and literary works

IP Laws Protect Creators

- IP laws provide **exclusive rights to the creator(s)** of an intellectual property (IP)

→ Give creators the ability to control how their creations can be used (for a definite or indefinite period of time)

→ To incentivize the creation of the IP in the first place

IP Laws Protect Creators

If you are an inventor or creator of original works, IP laws…

- ➢ Prevent others from claiming ownership of your idea
- ➢ Prevent others from using your works inappropriately
- ➢ Attract investors to fund your inventive or creative idea
- ➢ Allows you to generate future revenues by licensing your IP
  - Licensing: partial transfer of IP rights, and get royalty payments on the licenses

Why Are IP Laws Important?

- For creators
- For [private businesses]
- For society at large

IP Laws Are Important for <u>Businesses</u>

Large industrial conglomerates own a large amount of IP

- They often hold a portfolio of different IP types: patents, trademarks, etc.

Economy-wide importance of IP-intensive industries

- They account for 41% of U.S. economic activities, and >62 million jobs (44% of total employment)

Well-known brands (trademark) attract consumers and generate revenues because of its goodwill (established reputation)

- Of course, also: movie studios, record labels, art galleries, etc.

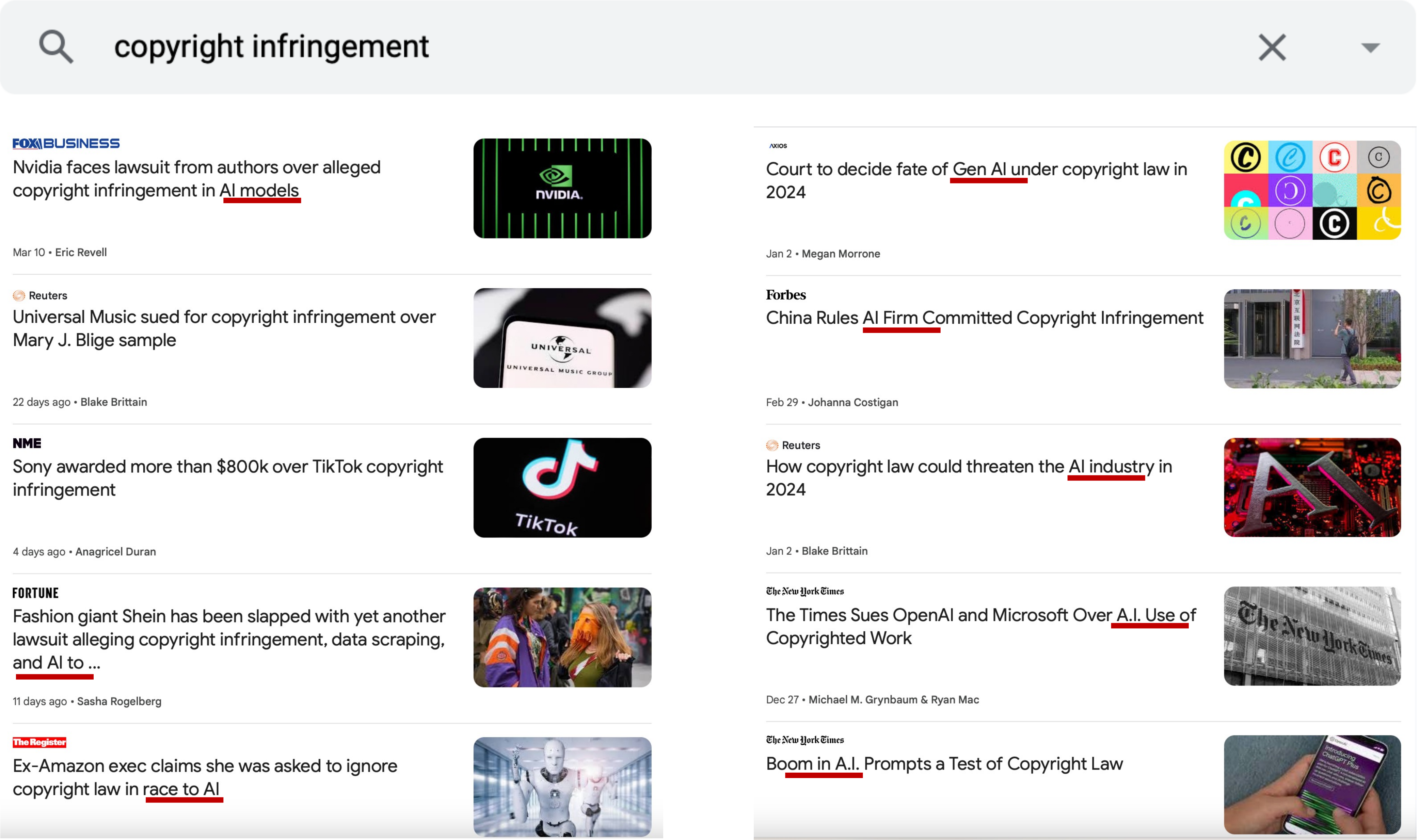
Why Are IP Laws Important?

- For creators
- For private businesses
- For society at large

IP Laws Are Important for Society

- IP laws aim to incentivize inventions and creative works – public goods that are valuable to the society

- Do they effectively achieve their intended goals?



copyright infringement

FOX BUSINESS
Nvidia faces lawsuit from authors over alleged copyright infringement in AI models
Mar 10 • Eric Revell

Reuters
Universal Music sued for copyright infringement over Mary J. Blige sample
22 days ago • Blake Brittain

NME
Sony awarded more than $800k over TikTok copyright infringement
4 days ago • Anagricel Duran

FORTUNE
Fashion giant Shein has been slapped with yet another lawsuit alleging copyright infringement, data scraping, and AI to ...
11 days ago • Sasha Rogelberg

The Register
Ex-Amazon exec claims she was asked to ignore copyright law in race to AI

AXIOS
Court to decide fate of Gen AI under copyright law in 2024
Jan 2 • Megan Morrone

Forbes
China Rules AI Firm Committed Copyright Infringement
Feb 29 • Johanna Costigan

Reuters
How copyright law could threaten the AI industry in 2024
Jan 2 • Blake Brittain

The New York Times
The Times Sues OpenAI and Microsoft Over A.I. Use of Copyrighted Work
Dec 27 • Michael M. Grynbaum & Ryan Mac

The New York Times
Boom in A.I. Prompts a Test of Copyright Law

Social Costs of IP Protection

- IP protection has costs and downsides, too…

→ It takes times and resources to *obtain* and *enforce* IP rights

→ *Litigations* can be very costly to businesses

→ It can slow down innovation as *entrepreneurs* need to avoid infringing on existing IP

A LOT OF academic literature…

How Do Patent Laws Influence Innovation? Evidence from Nineteenth-Century World's Fairs

Petra Moser

AMERICAN ECONOMIC REVIEW
VOL. 95, NO. 4, SEPTEMBER 2005
(pp. 1214–1236)

Download Full Text PDF

Article Information

Abstract

Studies of innovation have focused on the effects of patent laws on the number of innovations, but have ignored effects on the direction of technological change. This paper introduces a new dataset of close to fifteen thousand innovations at the Crystal Palace World's Fair in 1851 and at the Centennial Exhibition in 1876 to examine the effects of patent laws on the direction of innovation. The paper tests the following argument: if innovative activity is motivated by expected profits, and if the effectiveness of patent protection varies across industries, then innovation in countries without patent laws should focus on industries where alternative mechanisms to protect intellectual property are effective. Analyses of exhibition data for 12 countries in 1851 and 10 countries in 1876 indicate that inventors in countries without patent laws focused on a small set of industries where patents were less important, while innovation in countries with patent laws appears to be much more diversified. These findings suggest that patents help to determine the direction of technical change and that the adoption of patent laws in countries without such laws may alter existing patterns of comparative advantage across countries.

ANNUAL REVIEW OF ECONOMICS Volume 8, 2016

Review Article | Free

Patents and Innovation in Economic History

Petra Moser[1]
View Affiliations

Vol. 8:241-258 (Volume publication date October 2016) | https://doi.org/10.1146/annurev-economics-080315-015136

First published as a Review in Advance on August 08, 2016

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ABSTRACT

A strong tradition in economic history, which relies primarily on qualitative evidence and statistical correlations, has emphasized the importance of patents as a primary driver of innovation. Recent improvements in empirical methodology—through the creation of new datasets and advances in identification—have produced research that challenges this traditional view. The findings of this literature provide a more nuanced view of the effects of intellectual property and suggest that when patent rights have been too broad or too strong, they have actually discouraged innovation. This review summarizes the major results from this research and presents open questions.

…on the nuanced impact of IP protection

Social Costs of IP Protection

- **Market power** of IP owners: they may attempt to maintain market dominance and stifle competition

- Rent seeking behavior: “an entity seeks to gain wealth without reciprocal contribution of productivity”

Social Costs of IP Protection

- Focus on securing IP rights **distracts firms away from the innovation itself**

- Firms may spend too many resources or efforts to obtain IP rights (e.g., patent trolls) and protect their IP (e.g., litigations)

Social Costs of IP Protection

- When IP protection is too strong and broad, it can **constrain future innovation**
- If IP protection is used to prevent competitors' R&D efforts on similar technologies, it can hamper development of new technology that builds on prior inventions

Primary IP Types & Their Characteristics

	Who & What	How to obtain	Duration of protection
Patents	Original, novel inventions that are industrially applicable	File an application (may cost thousands of $) and be approved	Generally, 20 years from the patent application filing date
Copyright	Creative works of authorship, e.g., books, paintings, movies, songs, & computer software (considered literary work)	Automatically granted upon creation, but requires registration to be able to sue others for infringement	Person: author's life + 70 yrs Work for hire: 95 years from publication, or 120 years from creation (whichever comes first)
Trademarks	Distinctive symbols, names & images used in commerce that evokes association with a particular producer	Registration with the government	Indefinitely, until it loses distinctiveness or is no longer used in commerce

- International efforts to harmonize IP laws but still some differences across countries

Is There a Market for IP Rights?

- Intellectual Property (IP): a legal term referring to <u>intangible property that result from innovation and creativity</u>

- IP rights naturally belong to creators, but can be **<u>transferred</u>** via…

→ **<u>Assignment</u>**: transfer *all* rights of an IP

→ **<u>Licensing</u>**: transfer *partial* rights of an IP

IP Licensing Agreement

- Through licensing agreements, *IP owners can exercise certain rights* (e.g., buying, selling, licensing, leverage) in similar ways as those applied to physical properties

- *Firms* (not IP owners) can also use licensing agreements to *obtain intellectual property owned by others* for their businesses

IP Licensing Agreement

- The <u>**licensor**</u> (IP owner) retains ownership and control of the IP, while granting the <u>**licensee**</u> *certain rights*

There are many types of such rights

→ E.g., transfer the right to ***use*** the work, but ***prohibit third party from modifying*** the work

→ E.g., transfer the right to ***modify*** the work, but reserve the ***exclusive right to distribute*** the work to the public

Alternative to Copyright: Open-Source

Open-source software (OSS):

- Original **source code** that is designed to be *freely accessible* by the public, and anyone can *modify* and *redistribute* it

- Lower costs to follow-on development work, but require dedicated long-term maintenance

Open-Source Software (OSS)

Two types of OSS:

- Permissive (minimal restrictions) : can modify and redistribute freely, only minimal restrictions on how the software can be used
- Copyleft (requires reciprocity): free but require derivative work to be distributed under a similar license

Most Common OSS Licenses

License	Class	Visitors
mit	Permissive	1M
bsd-3-clause	Permissive	207K
apache-2-0	Permissive	184K
bsd-2-clause	Permissive	98.2K
gpl-2-0	Copyleft	71.7K
gpl-3-0	Copyleft	61.5K

Source: https://opensource.org/blog/top-open-source-licenses-in-2024

Comparing Common OSS Licenses

Permissive licenses maximize downstream flexibility

- **MIT/BSD:** widest adoption & minimal legal frictions
- **Apache 2.0:** almost same as MIT/BSD + patent grant (prevents contributors from suing users over patents)

Copyleft licenses maximizes enduring openness

- **GPL:** downstream applications and improvements must stay open, and users always have the source code

OSS Impact on Firms and Society

Home > Organization Science > Vol. 14, No. 2 >

Open Source Software and the "Private-Collective" Innovation Model: Issues for Organization Science

Eric von Hippel, Georg von Krogh

Published Online: 1 Apr 2003 | https://doi.org/10.1287/orsc.14.2.209.14992

Abstract

Currently, two models of innovation are prevalent in organization science. The "private investment" model assumes returns to the innovator result from private goods and efficient regimes of intellectual property protection. The "collective action" model assumes that under conditions of market failure, innovators collaborate in order to produce a public good. The phenomenon of open source software development shows that users program to solve their own as well as shared technical problems, and freely reveal their innovations without appropriating private returns from selling the software. In this paper, we propose that open source software development is an exemplar of a compound "private-collective" model of innovation that contains elements of both the private investment and the collective action models and can offer society the "best of both worlds" under many conditions. We describe a new set of research questions this model raises for scholars in organization science. We offer some details regarding the types of data available for open source projects in order to ease access for researchers who are unfamiliar with these, and also offer some advice on conducting empirical studies on open source software development processes.

Home > Management Science > Vol. 65, No. 3 >

Open Source Software and Firm Productivity

Frank Nagle

Published Online: 4 May 2018 | https://doi.org/10.1287/mnsc.2017.2977

Abstract

As open source software (OSS) is increasingly used as a key input by firms, understanding its impact on productivity becomes critical. This study measures the firm-level productivity impact of nonpecuniary (free) OSS and finds a positive and significant value-added return for firms that have an ecosystem of complementary capabilities. There is no such impact for firms without this ecosystem of complements. Dynamic panel analysis, instrumental variables, and a variety of robustness checks are used to address measurement error concerns and to add support for a more causal interpretation of the results. For firms with an ecosystem of complements, a 1% increase in the use of nonpecuniary OSS leads to an increase in value-added productivity of between 0.002% and 0.008%. This effect is smaller for larger firms, and the results indicate that prior research underestimates the amount of IT firms use.

Open-Source AI Development

Past OSS players: Apache, Linux, Mozilla, Google (e.g., TensorFlow, Chromium, Kubernetes), Meta (e.g., React, PyTorch, GraphQL)

Google and Meta played major role in open-source deep learning

- TensorFlow, Keras, JAX (Google) → production
- PyTorch (Meta) → research
- Pre-history of GPT-class LLMs: word2vec, Transformer, BERT (Google)

Open-Source AI Development

Recent trends: open-source large language models (LLMs)

- Meta (llama) & Google (gemma) among the leaders
- New prominent players including startups and non-US firms: e.g., Mistral AI, Stability AI, Deepseek, Alibaba (Qwen)
- Also, IBM (granite), Microsoft (Phi), BLOOM (BigScience / HuggingFace+collaborators), and many more

Open-Source Large Language Models

Questions:

- What are the main *differences* between open-source LLMs and closed proprietary LLMs?
- What are the *advantages* of open-source LLMs relative to closed and proprietary LLMs?
- What are some major *risks* of open-source LLMs?

New Types of Open-Source Licenses?

Typical OSS licenses may not fit open-source LLMs well

Generally, machine learning (ML) models are not exactly software

→ OSS protection is based on software source code written by humans, but ML models are not just source code

→ ML models are trained based on ***input data*** provided by humans

→ LLM ***"learns" complex patterns by itself*** (e.g., self-supervised learning) without human instructions

Licensing Types of Open-Source LLMs

Not all open-source LLMs are equally open…

- Some are permissive or copyleft (like typical OSS licenses)
- But many high-end LLMs are either dual-licensed or source-available (i.e., additional restrictions apply selectively)
- Charge users with commercial license when the scale or competitive use threatens the licensor's core business

Licensing Types of Open-Source LLMs

Licence Type	Characteristics	Examples
Permissive	OSI-approved	Mistral (small), Gemma, Phi, Granite
Copyleft	Requires reciprocity	GPT-J-6B, community fine-tuned models

→ These are the traditional OSS license types, which do not cover new types of licensing arrangements for open-source LLMs, e.g.:

- Dual-licensed and source-available LLMs

Licensing Types of Open-Source LLMs

Licence Type	Characteristics	Examples
Permissive	OSI-approved	Mistral (small), Gemma, Phi, Granite
Copyleft	Requires reciprocity	GPT-J-6B, community fine-tuned models
Dual-licensed	OSI + content & use-case restrictions	BLOOM, Qwen, Stable LM
Source-available	Weights visible and modifiable, but field-of-use, model-size, and user-base restrictions	Llama, Grok, Mistral (large)

Licensing Types of Open-Source LLMs

If you are a developer, an entrepreneur, or otherwise commercial user of open-source LLMs…

Practical considerations in choosing which LLMs to use:

- A more permissive license is better to avoid downstream legal troubles

Dual-Licensing: Additional Restrictions

- Field-of-use limits: Model weights cannot be used for certain industries or business models
- User-base caps: Above a certain number of monthly active users, a commercial license must be purchased
- Model-size or domain caps: Certain downstream model sizes or domains are prohibited without purchasing a commercial license

Dual-Licensing: Added Restrictions

Rationales for these added restrictions in dual-licensing

- **Free for most customers:** Permissive licensing for small players (e.g., researchers, amateur developers, small teams)
- **Monetizing high-value customers:** Commercial licensing for big players (e.g., enterprise large teams, professional developers)

→ The better the model the more likely it has added restrictions

Performance Evaluation of LLMs

Stanford Center for Research on Foundation Models (CRFM), Holistic Evaluation of Language Models (HELM)

Accuracy | Efficiency | General information

Model	Mean win rate	NarrativeQA - F1	NaturalQuestions (open) - F1	NaturalQuestions (closed) - F1	OpenbookQA - EM	MMLU - EM
GPT-4o (2024-05-13)	**0.938**	**0.804**	0.803	0.501	0.966	0.748
GPT-4o (2024-08-06)	0.928	0.795	0.793	0.496	0.968	0.738
DeepSeek v3	0.908	0.796	0.765	0.467	0.954	0.803
Claude 3.5 Sonnet (20240620)	0.885	0.746	0.749	**0.502**	**0.972**	0.799
Amazon Nova Pro	0.885	0.791	**0.829**	0.405	0.96	0.758
GPT-4 (0613)	0.867	0.768	0.79	0.457	0.96	0.735
GPT-4 Turbo (2024-04-09)	0.864	0.761	0.795	0.482	0.97	0.711
Llama 3.1 Instruct Turbo (405B)	0.854	0.749	0.756	0.456	0.94	0.759

Leaderboard: Core scenarios - accuracy

Performance Evaluation of LLMs

- Stanford HELM Leaderboard: https://crfm.stanford.edu/helm/lite/latest/#/leaderboard

- A few top open-source LLMs perform on par with proprietary LLMs: e.g., DeepSeek v3 and Llama 3.1 Instruct Turbo (405B)

Case: SB1047 (California, 2024)

SB 1047 is the failed 2024 California senate bill…

“Safe and Secure Innovation for Frontier Artificial Intelligence Models Act”

which aimed to "mitigate the risk of catastrophic harms from AI models”

e.g., mass harms caused by autonomous biological or cyber-weapons of extremely powerful AIs

Case: SB1047 (California, 2024)

- SB 1047 would apply to
  - AI companies doing business in California regardless of their HQ location
  - All models that cost more than $100 million to train
  - All models that used computing power greater than 10^{26} FLOPs to train

- Introduced in 02/2024, vetoed by Governor Gavin Newsom in 09/2024 due to “targeting only model size rather than concrete risks when deployed in high-risk environments”

Case: SB1047 (California, 2024)

Main problems with SB 1047: Curtails innovation and experimentation

- AI technology evolves rapidly requiring adaptability in regulation: e.g., model training efficiency improves quickly → the compute size cap will be irrelevant because same model capabilities can be achieved by less and less compute
- Too cumbersome for developers (e.g., safety protocols, red-teaming tests, 3rd-party audits), draining tremendous time and money → only very large labs can afford these costs, while early-stage startups and academics cannot
- Practically kills open-source LLMs because it imposes obligations on model developers → a model's original developer may be liable for risks imposed by downstream derivatives (i.e., others who built on the model)

Big Techs and Open-Source LLMs

- Why do large tech firms like Meta and Alphabet engage in open-source LLM development?

→ Meta's open-source strategy

- Building for their own use cases, but eco-system complementors fill in the gaps and build on their releases
- Focus on foundational general models, which can be adapted for different downstream tasks

→ "open weights", not open data or open training recipe

Source: https://stratechery.com/2025/an-interview-with-meta-ceo-mark-zuckerberg-about-ai-and-the-evolution-of-social-media/

Homework Assignment

Choose one of the following two options

- Present an in-depth analysis of an open-source LLM provider of your choice:
  - What are its best models, and what are the licensing types of these models? How can one use these models?
  - What are the advantages (and disadvantages) of this provider relative to its key competitors? (hint: many angles e.g., country regulation, GPU resources, financial resources, existing users)
- Analyse a copyright infringement lawsuit around Generative AI (next session):
  - Choose a case from DAIL: https://blogs.gwu.edu/law-eti/ai-litigation-database-search/
  - What is the current status of the lawsuit (list ALL recent updates)?
  - If the suit is not yet settled, do you think the plaintiff/defendant is going to win? Why?
  - If the suit is not yet settled, how will you rule as a judge? Why?

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